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TREND-BASED MOCK PAPER

GATE Instrumentation Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Instrumentation Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

20% of 200:

(Numerical Answer Type — Enter the numerical value)

Section B: Instrumentation Engineering

Q2. [1-NAT] [1 mark]

Determinant of 2x2 matrix:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-NAT] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q4. [2-MSQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q5. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q6. [2-MSQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \|u\| \|v\|$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q7. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q8. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q9. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q10. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q11. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q12. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q13. [2-MSQ] [2 marks]

The vector (1,2,3) projected onto (2,1,0) is:

- (A) (4/5,2/5,0)
- (B) (8/5,4/5,0)
- (C) (2,1,0)
- (D) (1,2,3)

Q14. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q15. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q16. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q18. [1-MCQ] [1 mark]

Mod-12 counter minimum FFs:

- (A) 3
- (B) 4
- (C) 12
- (D) 8

Q19. [1-NAT] [1 mark]

One's complement of 11010110:

(Numerical Answer Type — Enter the numerical value)

Q20. [1-MCQ] [1 mark]

4-bit Johnson counter states:

- (A) 4
- (B) 8
- (C) 16
- (D) 10

Q21. [2-MCQ] [2 marks]

4:1 MUX select lines:

- (A) 1

- (B) 2
- (C) 3
- (D) 4

Q22. [2-MCQ] [2 marks]

Race-around in JK FF when:

- (A) $J=K=0$
- (B) $J=K=1$, wide clock
- (C) $J=K=1$, short clock
- (D) $J \neq K$

Q23. [2-MCQ] [2 marks]

2K x 8 ROM address lines:

- (A) 2
- (B) 8
- (C) 11
- (D) 2048

Q24. [2-MCQ] [2 marks]

Setup time violation: solution:

- (A) Decrease clock freq
- (B) Increase clock freq
- (C) Use latch
- (D) Both A and C

Q25. [2-MCQ] [2 marks]

$F(A,B,C,D) = \sum m(0,1,2,3,4,5,8,9,10)$. Minimal:

- (A) $A' + B'C$
- (B) $A'D' + A'C'$
- (C) $B' + A'D'$
- (D) $A'B' + A'D'$

Q26. [2-MCQ] [2 marks]

4-bit ripple adder, 4 FAs at 20ns each. Delay:

- (A) 20ns
- (B) 40ns
- (C) 80ns
- (D) 100ns

Q27. [1-NAT] [1 mark]

Delta-Star transformation: R_a (star) =

(Numerical Answer Type — Enter the numerical value)

Q28. [1-NAT] [1 mark]

Two-port h-parameter h_{11} has units:

(Numerical Answer Type — Enter the numerical value)

Q29. [1-MCQ] [1 mark]

Superposition theorem applies to:

- (A) Linear only
- (B) Non-linear

- (C) Both
- (D) Time-varying only

Q30. [1-MCQ] [1 mark]

Thevenin equivalent with $V_{th}=10V$, $R_{th}=5\Omega$, load= 5Ω . Output:

- (A) 5V, 1A
- (B) 5V, 0.5A
- (C) 2.5V, 0.5A
- (D) 10V, 1A

Q31. [2-MCQ] [2 marks]

Norton equivalent of Thevenin ($V_{th}=10V$, $R_{th}=5\Omega$):

- (A) $I_n=2A$, $R_n=5$
- (B) $I_n=2A$, $R_n=10$
- (C) $I_n=0.5A$, $R_n=5$
- (D) $I_n=0.5A$, $R_n=10$

Q32. [2-MCQ] [2 marks]

$H(s) = (s^2+4s+3)/(s^2+2s+2)$. Zeros at:

- (A) $s=-1, -3$
- (B) $s=-1+j, -1-j$
- (C) $s=-1, -3$ and $-1+j$
- (D) $s=-2+j, -2-j$

Q33. [2-MCQ] [2 marks]

Time constant of RL ($R=10\Omega$, $L=5H$):

- (A) 2s
- (B) 0.5s
- (C) 50s
- (D) 0.05s

Q34. [2-NAT] [2 marks]

Y-parameter Y_{11} for series impedance Z is:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q36. [1-NAT] [1 mark]

PID controller has:

(Numerical Answer Type — Enter the numerical value)

Q37. [1-NAT] [1 mark]

Integral controller eliminates:

(Numerical Answer Type — Enter the numerical value)

Q38. [1-MCQ] [1 mark]

Nyquist criterion: $N = Z - P$. If $P=1$, $N=-1$:

- (A) 1 zero in RHP
- (B) Stable
- (C) 1 clockwise encirclement needed
- (D) Unstable

Q39. [1-NAT] [1 mark]

Root locus branches start at poles and end at:

(Numerical Answer Type — Enter the numerical value)

Q40. [1-MCQ] [1 mark]

Type 1 system with $K_v=5$. Steady-state error for unit ramp:

- (A) 0
- (B) 0.2
- (C) Infinity
- (D) 1

Q41. [2-MSQ] [2 marks]

Gain margin is defined at:

- (A) Phase crossover
- (B) Gain crossover
- (C) Both
- (D) Neither

Q42. [2-MCQ] [2 marks]

A system with characteristic $1+s = 0$ has:

- (A) Stable
- (B) Marginally stable
- (C) Unstable
- (D) Cannot determine

Q43. [2-MSQ] [2 marks]

$s^2 + 2s + 5 = 0$. Damping ratio:

- (A) 0.2
- (B) 0.447
- (C) 0.707
- (D) 1.0

Q44. [2-MCQ] [2 marks]

Phase margin is defined at:

- (A) Phase crossover
- (B) Gain crossover
- (C) Both
- (D) 0 dB crossing

Q45. [1-MCQ] [1 mark]

CE amp $h_{fe}=100$, $h_{ie}=1k$, $R_C=2k$. Voltage gain:

- (A) -200
- (B) -100
- (C) -150

(D) -50

Q46. [1-MCQ] [1 mark]

RC low-pass at cutoff, gain:

- (A) 0 dB
- (B) -3 dB
- (C) -20 dB
- (D) -40 dB/dec

Q47. [1-MCQ] [1 mark]

Op-amp integrator $R=10k$, $C=1\mu F$. Unit step after 1ms:

- (A) -0.1V
- (B) -0.01V
- (C) -0.001V
- (D) -1V

Q48. [2-MCQ] [2 marks]

Frequency response RC integrator (high-pass) corner:

- (A) $1/(2\pi RC)$
- (B) RC
- (C) $1/RC$
- (D) $2\pi RC$

Q49. [2-MCQ] [2 marks]

Class B push-pull efficiency:

- (A) 25%
- (B) 50%
- (C) 78.5%
- (D) 100%

Q50. [2-MSQ] [2 marks]

CC (emitter follower) has:

- (A) High voltage gain
- (B) High current gain
- (C) Low input Z
- (D) Voltage gain > 1

Q51. [1-MCQ] [1 mark]

Shannon capacity: $BW=1MHz$, $SNR=15$. Capacity:

- (A) 4 Mbps
- (B) 1 Mbps
- (C) 15 Mbps
- (D) 5 Mbps

Q52. [1-MCQ] [1 mark]

NBFM bandwidth approx:

- (A) $2f_m$
- (B) $2(\Delta f + f_m)$
- (C) Δf
- (D) f_m only

Q53. [1-MCQ] [1 mark]

AM signal carrier 1kW, modulation 0.5. Total power:

- (A) 1.125 kW
- (B) 1.5 kW
- (C) 1.0 kW
- (D) 0.5 kW

Q54. [2-MCQ] [2 marks]

QPSK transmits bits per symbol:

- (A) 1
- (B) 2
- (C) 4
- (D) 8

Q55. [2-MSQ] [2 marks]

Carson rule: $\Delta f = 75\text{kHz}$, $f_m = 15\text{kHz}$. BW:

- (A) 180 kHz
- (B) 90 kHz
- (C) 150 kHz
- (D) 160 kHz

Q56. [2-MCQ] [2 marks]

Nyquist rate for $f_{\text{max}} = 4\text{kHz}$:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 16kHz

Q57. [1-MCQ] [1 mark]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q58. [1-NAT] [1 mark]

Sum of first n odd numbers:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q60. [1-NAT] [1 mark]

Instrumentation amp has:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

8-bit ADC has resolution:

- (A) $1/256$
- (B) $1/128$
- (C) $1/512$
- (D) $1/64$

Q62. [1-MCQ] [1 mark]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q63. [1-NAT] [1 mark]

FM uses:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q65. [1-MCQ] [1 mark]

Moving coil instrument measures:

- (A) AC only
 - (B) DC only
 - (C) Both
 - (D) None
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1NAT	A	1
3	1MCQ	A	1
4	1NAT	C	1
5	1NAT	C	1
6	1MCQ	A	1
7	1MCQ	B	1
8	1NAT	B	1
9	1MCQ	B	1
10	1MCQ	A	1
11	1NAT	B	1
12	1NAT	B	1
13	1MCQ	B	1

14	1MCQ	C	1
15	1MCQ	A	1
16	1NAT	A	1
17	1MCQ	A	1
18	1MCQ	B	1
19	1MCQ	A	1
20	1MCQ	A	1
21	1MCQ	A	1
22	1MCQ	A	1
23	1NAT	C	1
24	1NAT	A	1
25	1MCQ	B	1
26	1NAT	A	1
27	1MCQ	A	1
28	1MCQ	B	1

29	1NAT	B	1
30	1MCQ	A	1
31	2MSQ	A	2
32	2MCQ	B	2
33	2MSQ	B	2
34	2MCQ	A	2
35	2MSQ	B	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	B	2
39	2MCQ	A	2
40	2MCQ	A	2
41	2MCQ	B	2
42	2MSQ	A	2
43	2MCQ	A	2

44	2MCQ	C	2
45	2MCQ	A	2
46	2MCQ	B	2
47	2MCQ	C	2
48	2MCQ	C	2
49	2NAT	D	2
50	2MCQ	A	2
51	2MCQ	C	2
52	2MCQ	B	2
53	2MCQ	C	2
54	2MSQ	B	2
55	2MCQ	B	2
56	2MCQ	B	2
57	2MSQ	B	2
58	2NAT	C	2

59	2NAT	A	2
60	2MCQ	A	2
61	2MCQ	B	2
62	2MSQ	A	2
63	2MCQ	A	2
64	2MCQ	B	2
65	2MCQ	C	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: easy

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Sensors & Instrumentation | Topic: Transducers | Difficulty: moderate

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Control Systems | Topic: Controllers | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Analog & Digital Electronics | Topic: Op-Amps | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Analog & Digital Electronics | Topic: ADC | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Communications & Process Control | Topic: Control Loops | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Communications & Process Control | Topic: Modulation | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Measurement Systems | Topic: Instrumentation | Difficulty: moderate

Q10.

Answer: A

Solution:

$A_v = -h_{fe} R_C / h_{ie} = -100 \times 2000 / 1000 = -200.$

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q11.

Answer: B

Solution:

Integral control: eliminates steady-state error for step/ramp inputs. Increases system type by 1.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q12.

Answer: B

Solution:

$$R_a = (R1 \cdot R2 + R2 \cdot R3 + R3 \cdot R1) / R3.$$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q13.

Answer: B

Solution:

At f_c : gain = $1/\sqrt{2}$ = -3 dB.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q14.

Answer: C

Solution:

$N = Z - P$. $N = -1$ means one clockwise encirclement of -1 point needed. $Z = N + P = -1 + 1 = 0$. Stable.

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q15.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Signal Processing | Topic: Signal Processing | Difficulty: moderate

Q16.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Signal Processing | Topic: Signal Processing | Difficulty: difficult

Q17.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Signal Processing | Topic: Signal Processing | Difficulty: moderate

Q18.

Answer: B

Solution:

$2^n \geq 12$. $n=4$ gives 16 states. Minimum 4 FFs.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: difficult

Q19.

Answer: A

Solution:

$$C = B \cdot \log_2(1+SNR) = 1e6 \cdot \log_2(16) = 1e6 \cdot 4 = 4 \text{ Mbps.}$$

Subject: Communication | Topic: Communication | Difficulty: moderate

Q20.

Answer: A

Solution:

$$V_{out} = -t/(RC) = -100 \cdot t. \text{ At 1ms: } -0.1V.$$

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q21.

Answer: A

Solution:

NBFM: BW approx $2 \cdot f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q22.

Answer: A

Solution:

$$P_{\text{total}} = P_c(1+m^2/2) = 1000*(1+0.125) = 1125W = 1.125kW.$$

Subject: Communication | Topic: Communication | Difficulty: easy

Q23.

Answer: C

Solution:

Root locus: starts at OL poles, ends at OL zeros (or infinity if $P > Z$).

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q24.

Answer: A

Solution:

Flip all bits: 11010110 -> 00101001.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q25.

Answer: B

Solution:

Johnson (4-bit): 8 states.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q26.

Answer: A

Solution:

$h_{11} = (V_1/I_1)$ with output shorted. Units: $V/A = \text{Ohm}$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q27.

Answer: A

Solution:

Superposition requires linearity (homogeneity + additivity). Does NOT apply to non-linear.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q28.

Answer: B

Solution:

Type 1: $ess = 1/K_v = 1/5 = 0.2$ for unit ramp.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q29.

Answer: B

Solution:

$$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

$$V_{\text{out}} = 10*5/(5+5) = 5V. I = 5/5 = 1A.$$

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q31.

Answer: A

Solution:

Gain margin: at phase crossover frequency (phase = -180 deg), $1/|GH(j\omega)|$ in dB.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q32.

Answer: B

Solution:

Root at $s = -1$ (on left half plane). Simple pole on negative real axis: stable (marginally, as $t \rightarrow \infty$).

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q33.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

$I_n = V_{th}/R_{th} = 10/5 = 2A$. $R_n = R_{th} = 5\Omega$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q35.

Answer: B

Solution:

Compare to $s^2 + 2*\zeta*\omega_n*s + \omega_n^2$. $2*\zeta*\sqrt{5} = 2$. $\zeta = 2/(2*\sqrt{5}) = 1/\sqrt{5}$ approx 0.447.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q36.

Answer: B

Solution:

QPSK: 4 phase states. $\log_2(4) = 2$ bits per symbol.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q37.

Answer: A

Solution:

Zeros from numerator: $s^2+4s+3 = (s+1)(s+3)$. Zeros at $s=-1, -3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: difficult

Q38.

Answer: B

Solution:

4:1: $\log_2(4) = 2$ select lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q39.

Answer: A

Solution:

RC high-pass cutoff: $1/(2*\pi*RC)$.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: difficult

Q40.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: B

Solution:

Race-around: $J=K=1$ with clock pulse wider than propagation delay.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: difficult

Q42.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: C

Solution:

$2K = 2048$ locations. $\log_2 = 11$ address lines.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q45.

Answer: A

Solution:

Setup: data not stable before clock. Lower clock freq gives longer period.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: easy

Q46.

Answer: B

Solution:

$\tau = L/R = 5/10 = 0.5$ s.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q47.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q48.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q49.

Answer: D

Solution:

Series Z: $I_1=I_2$, $V_1-V_2=Z \cdot I_1$. $Y_{11} = I_1/V_1$ with $I_2=0$ impossible ($I_1=I_2$). $Y_{11}=0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q50.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q52.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q53.

Answer: C

Solution:

Class B max efficiency: $\pi/4$ approx 78.5%.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q54.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q55.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q56.

Answer: B

Solution:

Phase margin: at gain crossover ($|GH| = 1$), $1 +$ phase in degrees.

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q57.

Answer: B

Solution:

CC: $A_v \sim 1$, high input Z, high current gain, low output Z.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q58.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q59.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: B

Solution:

K-map: $F = A'(D' + B'C') = A'D' + A'B'C'$. Closest: $A'D' + A'C'$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q62.

Answer: A

Solution:

$BW = 2 \times (75 + 15) = 180$ kHz.

Subject: Communication | Topic: Communication | Difficulty: moderate

Q63.

Answer: A

Solution:

$L\{e^{-(s+a)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q64.

Answer: B

Solution:

$f_s \geq 2 \times 4\text{kHz} = 8\text{kHz}$.

Subject: Communication | Topic: Communication | Difficulty: easy

Q65.

Answer: C

Solution:

Carry ripples through 4 FAs. $t_{pd} = 4 \times 20 = 80\text{ns}$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: difficult

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 16 questions, 30 marks
Digital Electronics: 9 questions, 15 marks
Network Theory: 9 questions, 14 marks
Control Systems: 9 questions, 13 marks
Analog Electronics: 6 questions, 9 marks
Communication: 6 questions, 9 marks
Signal Processing: 3 questions, 3 marks
Analog & Digital Electronics: 2 questions, 2 marks
Communications & Process Control: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Sensors & Instrumentation: 1 questions, 1 marks
Measurement Systems: 1 questions, 1 marks

Difficulty Distribution:

Easy: 16 | Moderate: 35 | Difficult: 14

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.